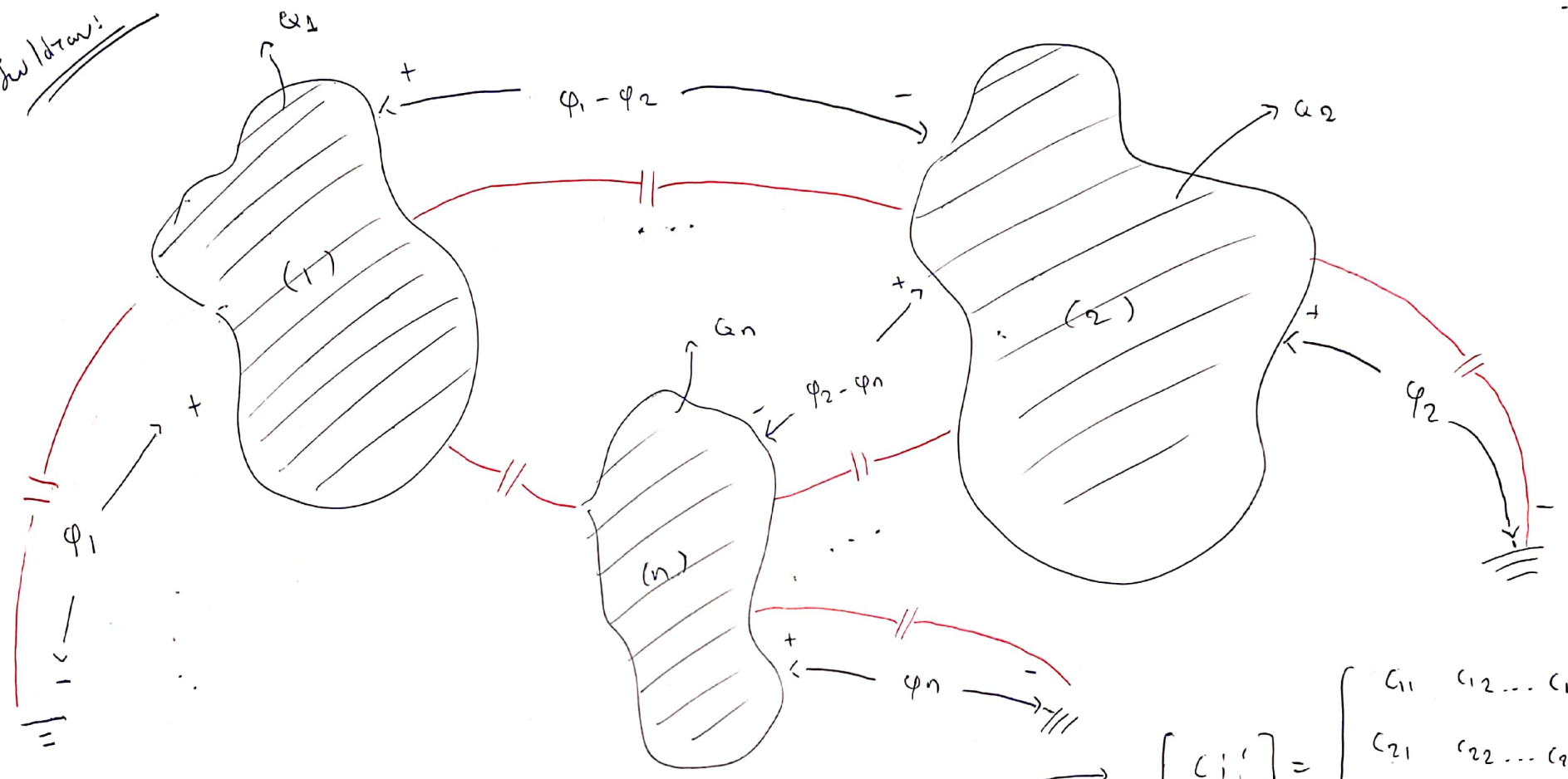


Συνολική
απόκριση κυκλώματος:



$$\begin{cases} Q_1 = c_{11}\phi_1 + c_{12}\phi_2 + \dots + c_{1n}\phi_n \\ Q_2 = c_{21}\phi_1 + c_{22}\phi_2 + \dots + c_{2n}\phi_n \\ \vdots \\ Q_n = c_{n1}\phi_1 + c_{n2}\phi_2 + \dots + c_{nn}\phi_n \end{cases}$$

$$\vec{Q} = \begin{bmatrix} c_{ij} \end{bmatrix} \vec{\phi} \quad \left\{ \begin{array}{l} \vec{\phi} = (\phi_1 \dots \phi_n)^T \\ \vec{Q} = (Q_1 \dots Q_n)^T \end{array} \right. \quad \left[c_{ij} \right] = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{bmatrix}$$

Συνολική Αμοιβαιότητα

$c_{ij} \sim$ συντελεστής χωρητικότητας

$$\begin{cases} c_{ij} > 0 \\ c_{ij} = c_{ji} < 0 \end{cases}$$

Σαρώθηκε με το CamScanner

$$[c_{ij}]^{-1} \bar{Q} = [c_{ij}]^{-1} [c_{ij}] \cdot \bar{\varphi} \xrightarrow{I} \bar{\varphi} = [c_{ij}]^{-1} \bar{Q} \rightsquigarrow \bar{\varphi} = [p_{ij}] \bar{Q}$$

Αντικ,

$$Q_1 = C_{11} \varphi_1 + C_{12} (\varphi_1 - \varphi_2) + \dots + C_{1n} (\varphi_1 - \varphi_n)$$

$$Q_2 = C_{21} (\varphi_2 - \varphi_1) + C_{22} \varphi_2 + \dots + C_{2n} (\varphi_2 - \varphi_n)$$

$$\vdots$$

$$Q_n = C_{n1} (\varphi_n - \varphi_1) + C_{n2} (\varphi_n - \varphi_2) + \dots + C_{nn} \cdot \varphi_n$$

$$[p_{ij}] = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & \dots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \dots & p_{nn} \end{bmatrix}$$

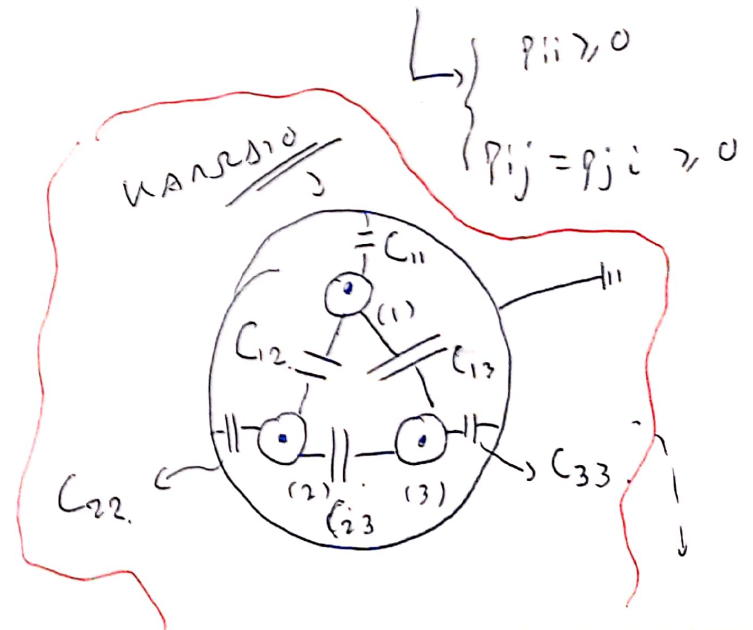
$C_{ij} \rightsquigarrow$ ΜΕΡΙΚΕΣ ΧΩΡΟΠΥΚΝΩΣΕΙΣ ΤΩ ΙΣΟΛΟΓΙΣΤΩ $\rightarrow$

ΙΣΟΛΟΓΙΣΤΗΣ ΔΥΝΑΜΙΔΟΥ

$$C_{ij} (= C_{ji}) = -c_{ij}$$

$$\rightarrow C_{ii} = C_{i1} + C_{i2} + \dots + C_{in} = \sum_{j=1}^n c_{ij}$$

$$C_{ij} \geq 0, \quad \forall (i,j)$$



Πρόβλημα
Αντιβασιότητας
του Green:

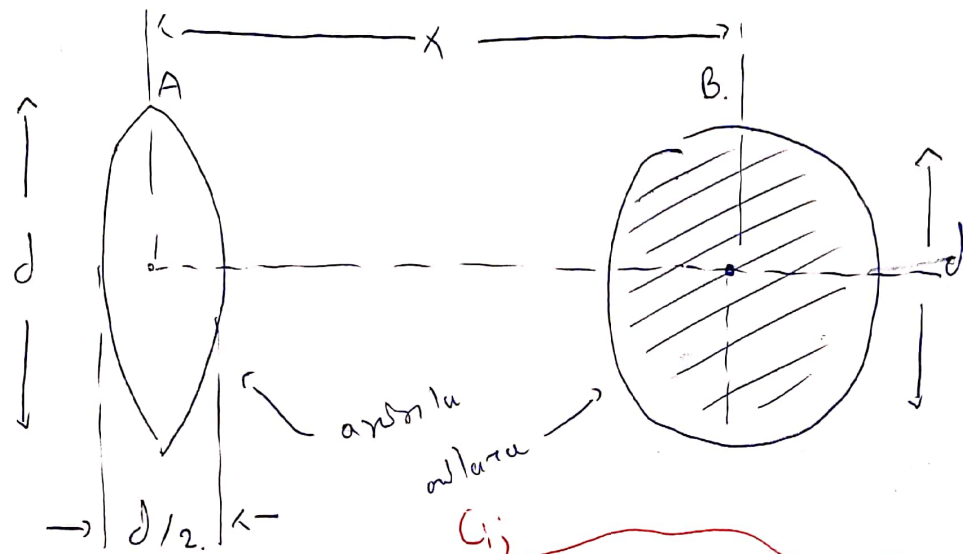
$$\sum_{i=1}^n \varphi_i' \cdot \alpha_i = \sum_{i=1}^n \varphi_i \cdot \alpha_i'$$

$$\varphi_1' \alpha_1 + \varphi_2' \alpha_2 + \dots + \varphi_n' \alpha_n =$$

$$= \varphi_1 \alpha_1' + \varphi_2 \alpha_2' + \dots + \varphi_n \alpha_n'$$

-3-
Λύνει τα φορτία ή τα δυναμικά
των γραμμών του δικτύου σε δύο διαφο-
ρετικές καταστάσεις κατά της τω-
ρας.

Lenig
(2):



(α) $\varphi_{A \kappa.3} = ?$

Έχω ένα αλάντα 2
αγωγιμότητας συνδεδεμένων, άρα.

Θα έχω

τα c_{ij} είναι
capacitances του συνδυασμού,
ΑΡΑ μένουν ίδιες
and variations
of variations.

κ. 1

$$\begin{cases} \varphi_A = c_{11} \varphi_A + c_{12} \varphi_B \\ \varphi_B = c_{21} \varphi_A + c_{22} \varphi_B \end{cases}$$

$c_{21} = c_{12}$

$$10 \cdot 10^{-6} = c_{11} \cdot 5 + c_{12} \cdot 0$$

$$\varphi_B = c_{12} \cdot 5 + c_{22} \cdot 0 \rightarrow$$

$$\rightarrow \varphi_B = 5 c_{12}$$

↓
δε μπορεί να
υπόκειται εδω
δε είναι ούτε c_{12}
ούτε φ_B .

καρατόμ 1

$$\varphi_B = 0, \varphi_A = 10 \mu C, \varphi_A = 5(V)$$

το B έχει φορτίο, αλλά δε μας γίει,
δε ν $\left\{ \begin{array}{l} \text{Α} \\ \text{Β} \end{array} \right\}$ (σπουδ νω).

κ. 2

$$\varphi_A = 0, \varphi_B = 9 \mu C, \varphi_B = 3(V)$$

κ. 3

$$\varphi_A = 2 \mu C, \varphi_B = 14 \mu C, \varphi_B = 6(V)$$

$$c_{11} = 2 \cdot 10^{-6} = 2 \mu F$$

κ. 2

$$\begin{cases} \varphi_A = c_{11} \cdot \varphi_A + c_{12} \cdot \varphi_B \\ \varphi_B = c_{21} \cdot \varphi_A + c_{22} \cdot \varphi_B \end{cases}$$

$$\begin{cases} \varphi_A = 3 c_{12} \\ c_{22} = 3 \cdot 10^{-6} = 3 \mu F \end{cases}$$

κ.3

$$\begin{cases} q_A = c_{11} \varphi_A + c_{12} \varphi_B \\ q_B = c_{21} \varphi_A + c_{22} \varphi_B \end{cases} \rightarrow \begin{cases} 2 \cdot 10^{-6} = 2 \cdot 10^{-6} \cdot \varphi_A + c_{12} \cdot 6 \rightarrow c_{12} = \frac{2 \cdot 10^{-6} - 2 \cdot 10^{-6} \varphi_A}{6} \\ 14 \cdot 10^{-6} = c_{12} \cdot \varphi_A + 3 \cdot 10^{-6} \cdot 6 \rightarrow 14 \cdot 10^{-6} = \frac{2 \cdot 10^{-6} - 2 \cdot 10^{-6} \varphi_A}{6} \cdot \varphi_A + 18 \cdot 10^{-6} \end{cases}$$

(β) $p_{ij}, c_{ij} = ?$

$$[c_{ij}] = \begin{bmatrix} c_{11} & c_{12} \\ c_{12} & c_{22} \end{bmatrix} =$$

$$\sim \frac{12}{24 \cdot 10^{-6}} + \frac{2}{10^{-6}} \varphi_A - \frac{2}{10^{-6}} \varphi_A^2 = 0 \rightarrow$$

$[p_{ij}] = [c_{ij}]^{-1}$
 $\hat{=}$
 $[c_{ij}] = [p_{ij}]^{-1}$

αντιστροφή

$$= \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}, (PF)$$

$$|c_{ij}| = 5$$

$$\rightarrow \varphi_A^2 - \varphi_A - 12 = 0 \rightarrow (\varphi_A - 4)(\varphi_A + 3) = 0$$

$\varphi_A = 4$ $\rightarrow$ $c_{12} = -1 (PF)$ ✓

$$\varphi_A = -3 \rightarrow c_{12} = \frac{8}{6} (PF) > 0$$

$$[p_{ij}] = \begin{bmatrix} 3 & +1 \\ +1 & 2 \end{bmatrix} \frac{1}{5} \rightarrow$$

$$\rightarrow [p_{ij}] = \begin{bmatrix} 3/5 & 1/5 \\ 1/5 & 2/5 \end{bmatrix}, p_{ij} > 0, \forall i, j$$

$$A = \begin{pmatrix} a & b \\ \gamma & \delta \end{pmatrix} \rightarrow$$

$$\rightarrow A^{-1} = \frac{1}{a\delta - b\gamma} \begin{pmatrix} \delta & -b \\ -\gamma & a \end{pmatrix}$$

Μερίδα χωρητικότητας

$$C_{11} = C_{11} + C_{12} = 2 - 1 = 1 \text{ (pF)}$$

$$C_{12} = -C_{12} = 1 \text{ (pF)}$$

οτι $C_{ij} > 0 \forall i, j$ οτι

$$C_{21} = -C_{21} = 1 \text{ (pF)}$$

$$C_{22} = C_{21} + C_{22} = -1 + 3 = 2 \text{ (pF)}$$

(γ) (11.30, κτ. 7 τωδόντο)

$$W_{x,3} = ?$$

$$W = \frac{1}{2} \sum_{i=1}^n Q_i \cdot V_i$$

$$= \frac{1}{2} Q_1 \cdot V_1 + \frac{1}{2} Q_2 \cdot V_2 = \frac{1}{2} q_{Au,3} \cdot \varphi_{Au,3} + \frac{1}{2} q_{Bu,3} \cdot \varphi_{Bu,3} =$$

(δ). Για φ_A, φ_B σταθερά, n ελφρα τω ρελιου
δη δμτω ωδ

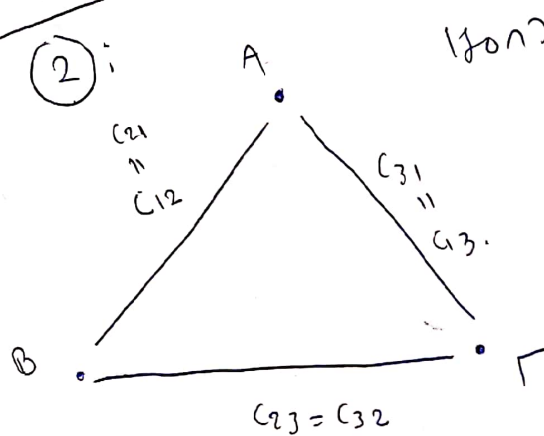
$$W_e = a + b d^2 \left(x - \frac{3d}{4} \right)^{-1} = \frac{1}{2} \cdot 2 \cdot 10^{-6} \cdot 4 + \frac{1}{2} \cdot 14 \cdot 10^{-6} \cdot 6 = 46 \text{ (kJ)}$$

(11.40, κτ. 8 τωδόντο)

$$L_1 F = \frac{\partial W_e}{\partial x}$$

$$\rightarrow F = \frac{9}{\partial x} \left(a + b d^2 \frac{1}{x - \frac{3d}{4}} \right) = - \frac{b d^2}{\left(x - \frac{3d}{4} \right)^2} \rightarrow F(x) \Big|_{x = \frac{43d}{4}} = - \frac{b d^2}{100 d^2} = - \frac{b}{100} \text{ (N)}$$

$\varphi \in \mathbb{R}^3$
(2):



ισογώνιο
τρίγωνο

→ ισόπλευρη συμμετρία
 $c_{12} = c_{13} = c_{23} = c_{\alpha}$

κ.1 $q_A = 0, q_B = q, \varphi_A = \varphi_1, \varphi_B = \varphi_2, \varphi_{\Gamma} = \varphi_2$

κ.2 $q_A = q', q_B = q'', q_{\Gamma} = q'', \varphi_A' = \varphi, \varphi_B' = 0, \varphi_{\Gamma}' = 0$

$q_{\Gamma} = ?; q'', q''' = ?$

$c_{\alpha} = -\frac{q' \varphi_1}{2 \varphi_2 \varphi}$

κ.1

$$\begin{cases} q_A = c_{11} \varphi_A + c_{12} \varphi_B + c_{13} \varphi_{\Gamma} \\ q_B = c_{21} \varphi_A + c_{22} \varphi_B + c_{23} \varphi_{\Gamma} \\ q_{\Gamma} = c_{31} \varphi_A + c_{32} \varphi_B + c_{33} \varphi_{\Gamma} \end{cases}$$

$$\begin{cases} 0 = c_{11} \varphi_1 + c_{\alpha} \varphi_2 + c_{\alpha} \varphi_2 \rightarrow \frac{q'}{\varphi} \varphi_1 + 2c_{\alpha} \varphi_2 = 0 \\ q = c_{\alpha} \varphi_1 + c_{22} \varphi_2 + c_{\alpha} \varphi_2 \rightarrow \frac{q - c_{\alpha} (\varphi_1 + \varphi_2)}{\varphi_2} = c_{22} \\ q_{\Gamma} = c_{\alpha} \varphi_1 + c_{\alpha} \varphi_2 + c_{33} \varphi_2 \end{cases}$$

κ.2

$$\begin{cases} q_A = c_{11} \varphi_A + c_{12} \varphi_B + c_{13} \varphi_{\Gamma} \\ q_B = c_{21} \varphi_A + c_{22} \varphi_B + c_{23} \varphi_{\Gamma} \\ q_{\Gamma} = c_{31} \varphi_A + c_{32} \varphi_B + c_{33} \varphi_{\Gamma} \end{cases}$$

$$\begin{cases} q' = c_{11} \cdot \varphi + \cancel{c_{\alpha} \cdot 0} + \cancel{c_{\alpha} \cdot 0} \rightarrow c_{11} = \frac{q'}{\varphi} \\ q'' = c_{\alpha} \cdot \varphi + \cancel{c_{22} \cdot 0} + \cancel{c_{\alpha} \cdot 0} \rightarrow q'' = c_{\alpha} \varphi \\ q''' = c_{\alpha} \cdot \varphi + \cancel{c_{\alpha} \cdot 0} + \cancel{c_{33} \cdot 0} \rightarrow q''' = c_{\alpha} \varphi \end{cases} \quad \left. \begin{matrix} q'' = c_{\alpha} \varphi \\ q''' = c_{\alpha} \varphi \end{matrix} \right\} q'' = q'''$$

ΠΩΣ
ΥΠΟΛΟΓΙΖΕΤΑΙ
ΤΟ q_5 ??

ο τωρ είναι η 2^η
υπόθεση, το αρθρ
η 2^η

$$C_{22} = \frac{q - \left(-\frac{q' \phi_1}{2\phi_2 \phi}\right) (\phi_1 + \phi_2)}{\phi_2} =$$

$$C_{22} = \frac{q}{\phi_2} + \frac{q' \phi_1}{2\phi_2^2 \phi} (\phi_1 + \phi_2)$$

Αν έχουμε

G. Green: $\sum_{i=1}^3 \phi_i' \omega_i = \sum_{i=1}^3 \phi_i \omega_i' \rightarrow$

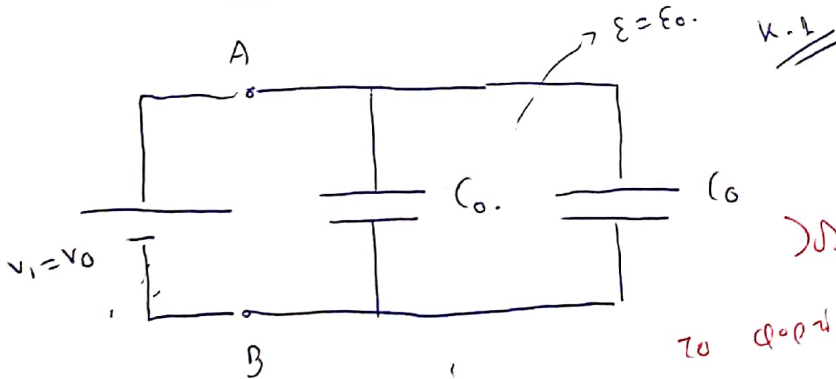
$$\rightarrow \phi_A' \omega_A + \phi_B' \omega_B + \phi_\Gamma' \omega_\Gamma = \phi_A \omega_A' + \phi_B \omega_B' + \phi_\Gamma \omega_\Gamma' \rightarrow \cancel{\phi} q_A + 0 \cdot \cancel{q_B} + 0 \cdot \cancel{q_\Gamma} = \phi_1 q' +$$

$$+ \phi_2 q'' + \phi_2 q''' \sim$$

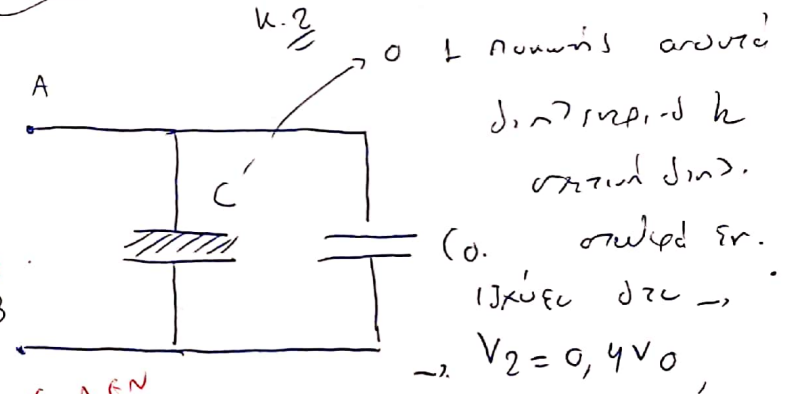
$$\sim \cancel{\phi} q_A = \phi_1 q' + \phi_2 q'' + \phi_2 q''' \sim \phi_1 q' = -2\phi_2 \phi$$

18/01/18
③:

(α)



Δίνω Α.Δ. Φορτίου
το φορτίο στο κ.1 ίδιο Β
η το φορτίο στο κ.2 (αλ C ΔΕΝ
έχω να εξοφλησω αλλιώς το
φορτίο των, άρα
ξέρω να εξοφλησω αλλιώς το
φορτίο των, άρα
ξέρω να εξοφλησω αλλιώς το
φορτίο των, άρα



στο ω_m α
ή το ίδιο?

Με το να συνδεθούν τα 2 πυκνωτές, αφού ο πυκνωτής (από ελάχιστη ενέργεια, αφού ο πυκνωτής) $Q_0 = C_0 \cdot V_0$

Αρα $Q_0 = Q_1 + Q_2 = 2 C_0 V_0, (C)$

• Για να () α ισχύει $C = \epsilon_r C_0$
(α: σταθερά διηλεκτρικού)

$$C = \epsilon_r \cdot \epsilon_0 \cdot \frac{A}{d}$$

$$\frac{C}{C_0} = \frac{\epsilon_r \cdot \epsilon_0 \cdot \frac{A}{d}}{\epsilon_0 \cdot \frac{A}{d}} \Rightarrow \frac{C}{C_0} = \epsilon_r \rightarrow C = \epsilon_r C_0$$

Συνολικό φορτίο με $V_2 \rightarrow Q_{02} = C \cdot V_2 + C_0 V_2 =$

$$= \epsilon_r C_0 \cdot V_2 + C_0 V_2 = C_0 \cdot 0,4 V_0 (\epsilon_r + 1)$$

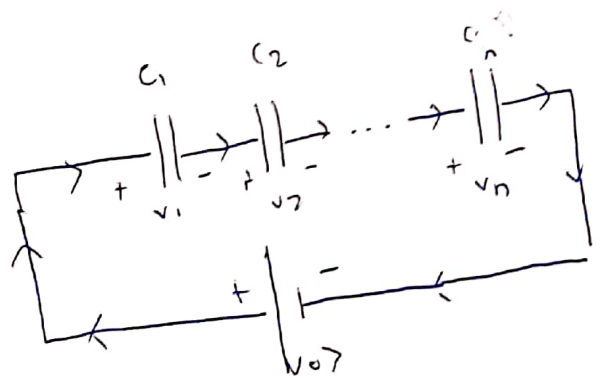
$$Q_{01} = Q_2 \rightarrow 2 C_0 V_0 = C_0 V_0 0,4 (\epsilon_r + 1) \rightarrow$$

$$\rightarrow \frac{2}{0,4} = \epsilon_r + 1 \rightarrow \boxed{\epsilon_r = 4}$$

Η σειρά χωρητήτων

$$\frac{1}{C_0} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n} = \sum_{i=1}^n \frac{1}{C_i}$$

$$Q_{01} = Q_1 = Q_2 = \dots = Q_n$$



$$i_{C1} = i_{C2} = \dots = i_{Cn} \rightarrow$$

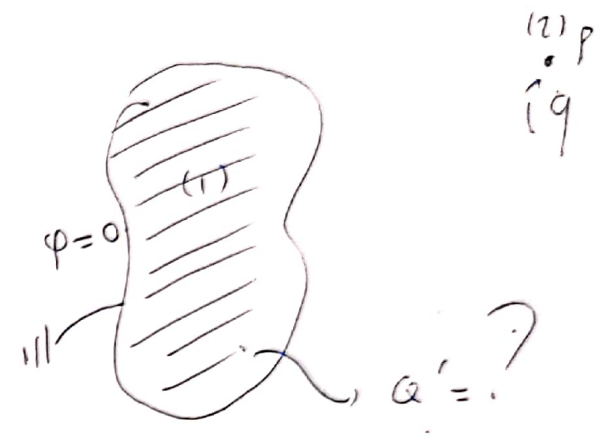
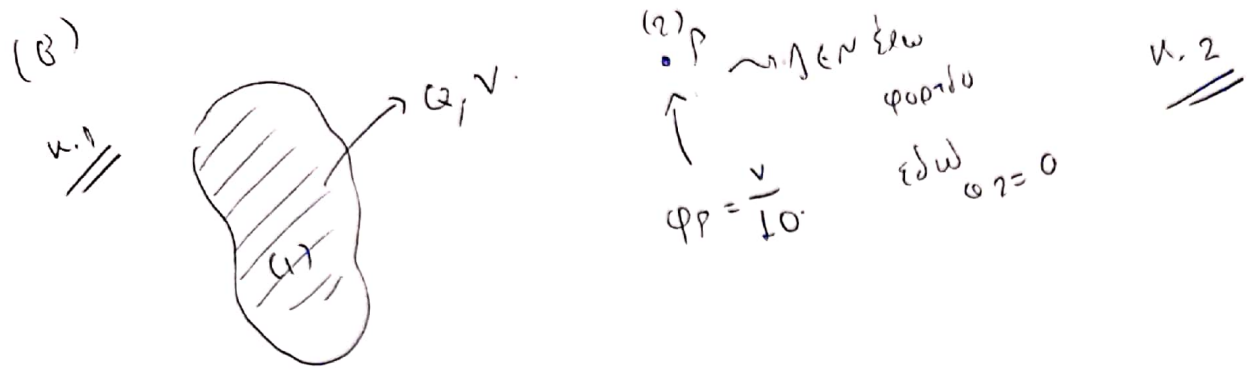
$$\frac{dQ_1}{dt} = \frac{dQ_2}{dt} = \dots = \frac{dQ_n}{dt} \rightarrow$$

$$\rightarrow Q_1 = Q_2 = \dots = Q_n$$

$V_0 = V_1 + V_2 + \dots + V_n \rightarrow$

$$\frac{Q_0}{C_0} = \frac{Q_1}{C_1} + \dots + \frac{Q_n}{C_n} \rightarrow$$

$$\rightarrow \frac{1}{C_0} = \sum_{i=1}^n \frac{1}{C_i}$$



9. Αποκρίσεις
Green

$$\sum_{i=1}^2 \varphi_i' Q_i = \sum_{i=1}^2 \varphi_i' Q_i' \rightarrow$$
$$\rightarrow \varphi_1' Q_1 + \varphi_2' Q_2 = \varphi_1' Q_1' + \varphi_2' Q_2' \rightarrow$$
$$\rightarrow \cancel{\varphi_1' Q_1} + \varphi_2' \cdot 0 = v Q' + \frac{v}{10} \cdot q \sim$$
$$\sim v Q' + \frac{v}{10} q = 0 \rightarrow$$
$$\rightarrow \frac{q}{10} = -Q' \sim$$

100N13

(1), (2), (4) ΕΙΕΙΣ

φΕΒ12

(1), (2), (3)

$Q' = -\frac{q}{10}$